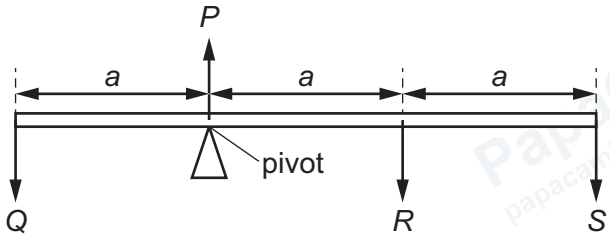
5 Four forces, *P*, *Q*, *R* and *S*, act on a wooden ruler as shown.



The ruler is pivoted so that it is in equilibrium. Ignore the weight of the ruler.

Which row is correct?

	forces	turning effects
A	$P = Q + R + S$	$(Q \times a) = (R \times a) + (S \times a)$
B	$P = Q + R + S$	$(Q \times a) = (R \times a) + (S \times 2a)$
C	$Q = R + S$	$(Q \times a) = (R \times a) + (S \times a)$
D	$Q = R + S$	$(Q \times a) = (R \times a) + (S \times 2a)$